Global Knowledge®

Angular Fundamentals

Routing

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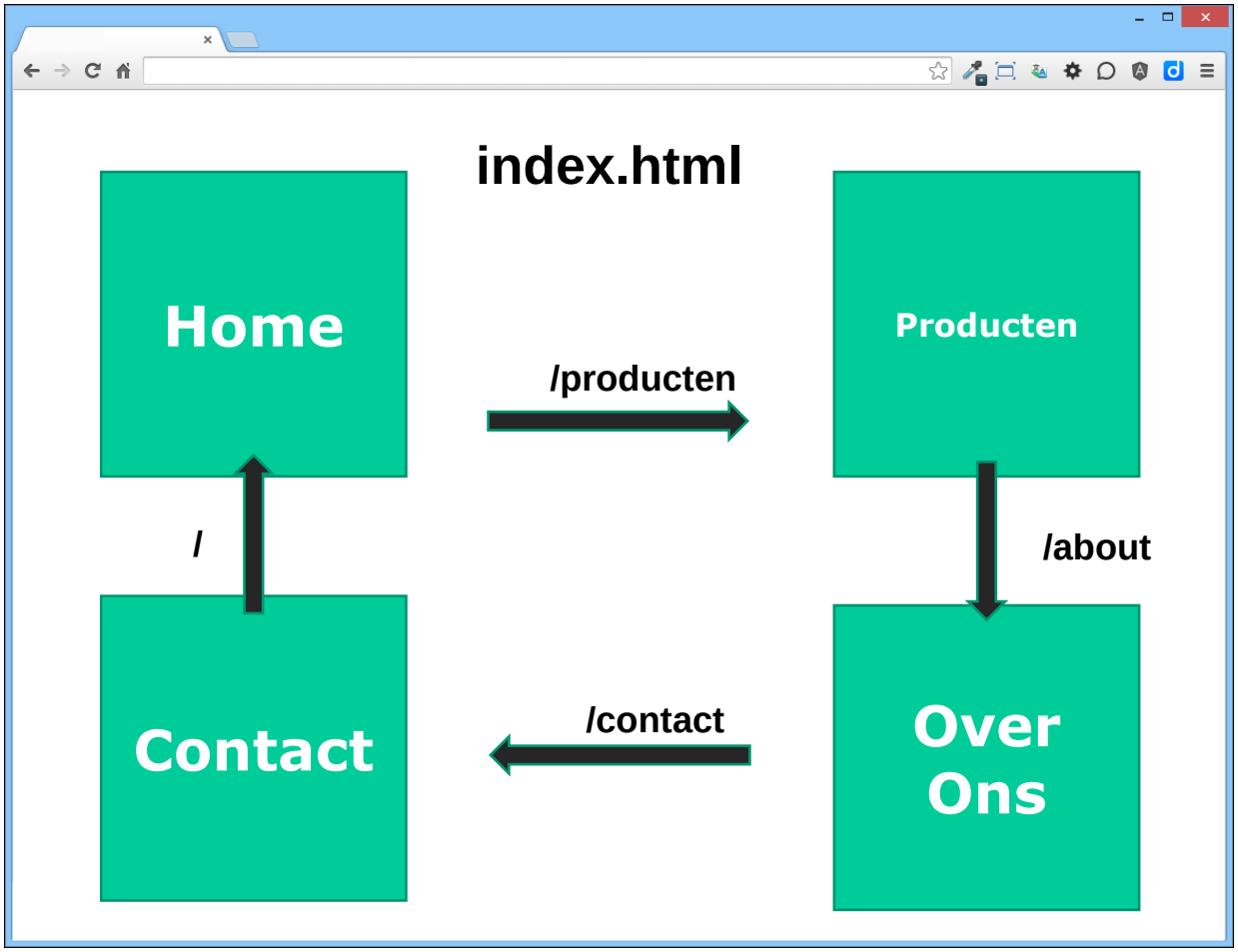
Routing architecture and goal



- Make use of SPA principle
- Making deep links possible

Angular 1: ng-route, of ui-router

```
1. <script src="js/vendor/angular/angular-route.min.js"></script>

2. <div ng-view></div>

3. var app = angular.module('myApp', ['ngRoute']);
```

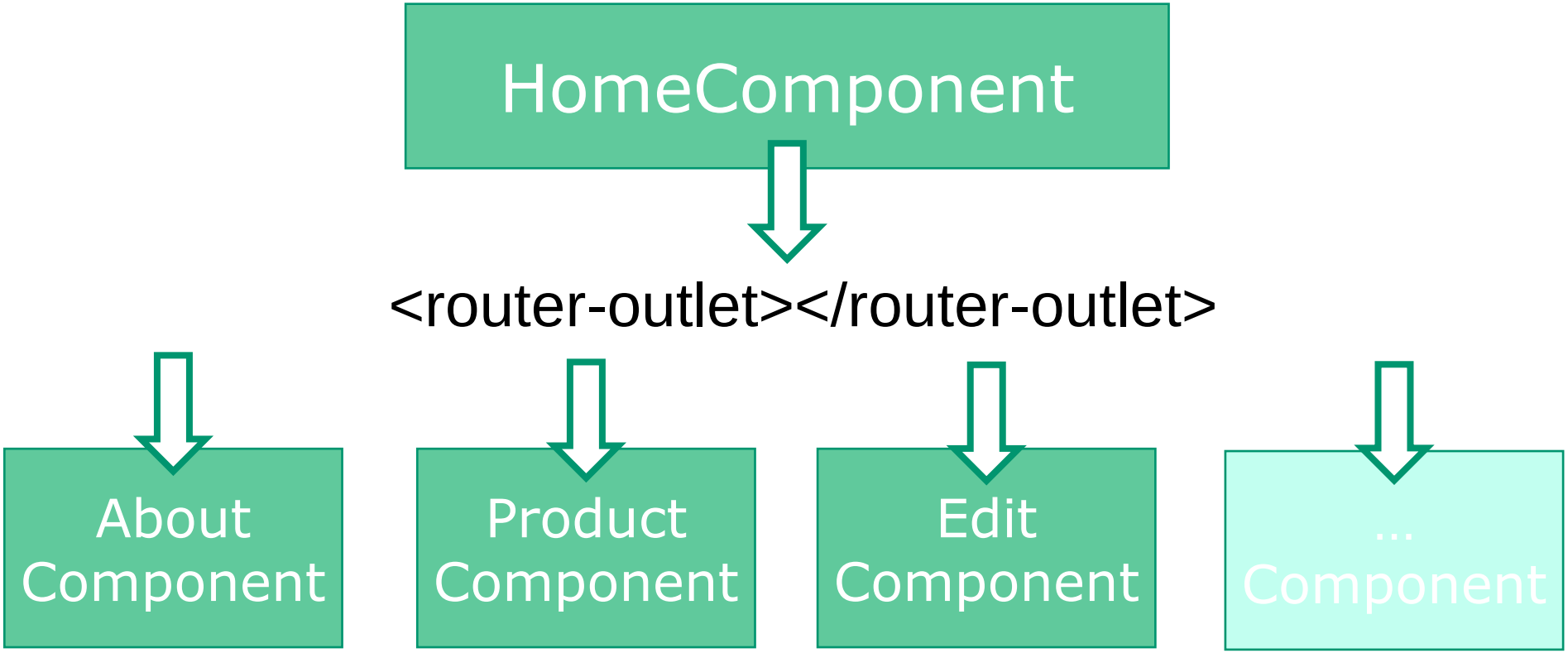
Daarna `$routeProvider` configureren (of `$stateProvider` bij ui-router)

Angular 2: Component Router

- Niet beschikbaar voor AngularJS 1.4+
- Niet veel gebruikt: ui-router

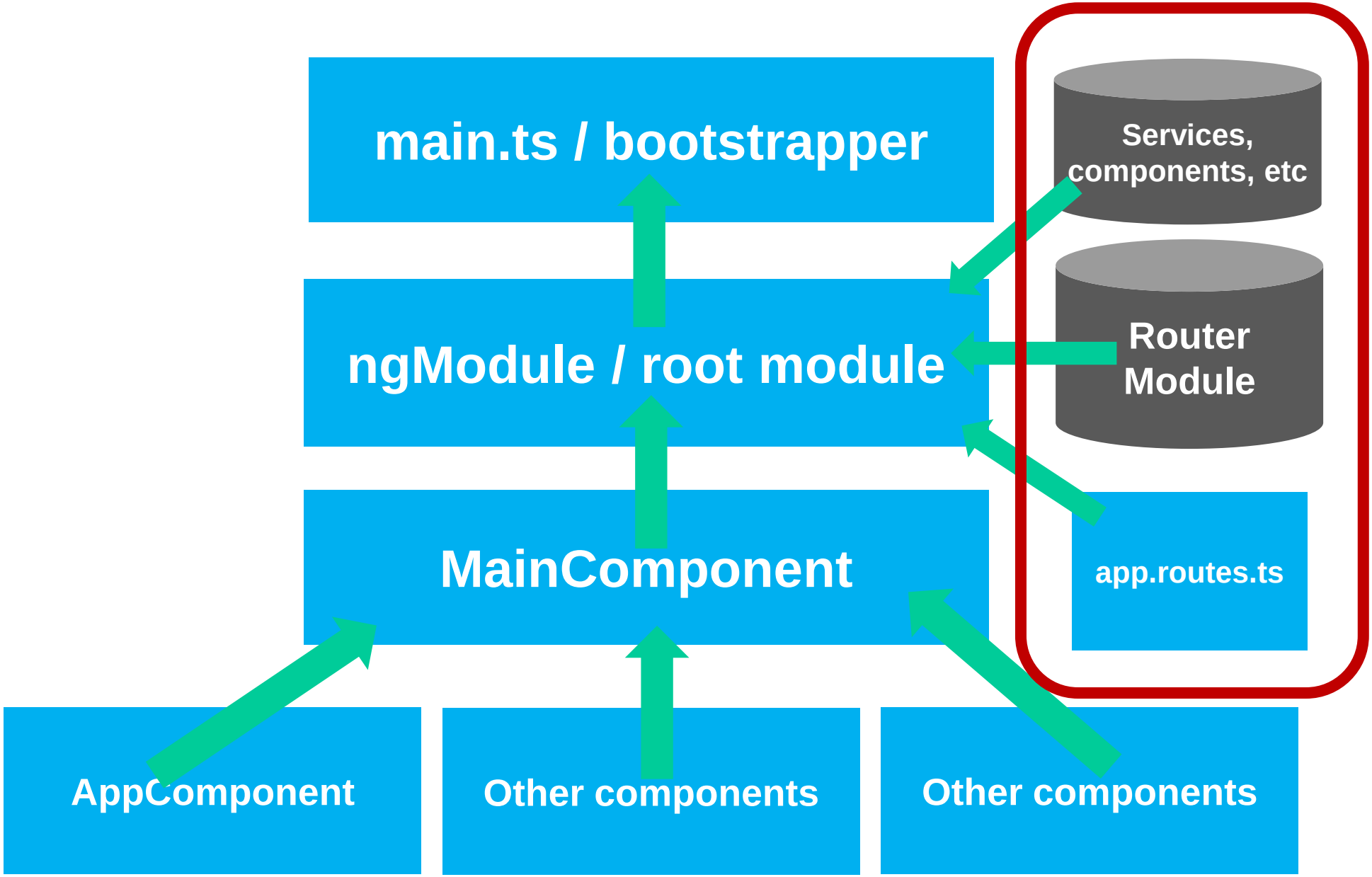
Routing – every route is a Component

- HomeComponent (or: RootComponent, whatever) with main menu
- Components are injected in `<router-outlet></router-outlet>`



Routing met Angular CLI < 17

- Standaard: geen routing in CLI-project
- Routing vanaf het begin toevoegen?
  - `ng new myProject --routing`
  - OF: vraag in CLI-command prompt met `Yes` beantwoorden
- Dit maakt `app.routing.module` in het project
- (iets) anders van opbouw dan we hier presenteren


main.ts / bootstrapper
Services, components, etc
ngModule / root module
Router Module
MainComponent
app.routes.ts
AppComponent
Other components
Other components

Stappenplan routing

1. Base Href toevoegen in header van index.html (!)

```
<base href="/">
```

- Er *kunnen* meerdere routes per module zijn. Elke component kan zijn eigen `ChildRoutes` definiëren – volgt later.
- Angular-CLI doet dit automatisch voor je.

2. Routes toevoegen. Convention: `app.routes.ts`.

```
// app.routes.ts
import {Routes} from '@angular/router';
import {AppComponent} from "./app.component";
import {CityAddComponent} from "./city.add.component";

export const AppRoutes: Routes = [
   {path: '', component: AppComponent},
   {path: 'home', component: AppComponent},
   {path: 'add', component: CityAddComponent}
];
```

Er zijn meerdere opties en notatiewijzen om routes te declareren

3. Routes beschikbaar maken in Module

- Import `RouterModule` in applicatie
- Import `./app.routes` in applicatie

```
…
// Router
import {RouterModule} from '@angular/router';
import {AppRoutes} from './app.routes';

// Components
import {MainComponent}  from './MainComponent';
…
@NgModule({
   imports     : [
      BrowserModule, HttpClientModule,
      RouterModule.forRoot(AppRoutes)
   ],
   declarations: [
      MainComponent,
      AppComponent,
      CityAddComponent
   ],
   bootstrap   : [MainComponent]
})
export class AppModule {
}
```

Import Router-onderdelen

Nieuw! MainComponent gaan we nog maken

Configure RouterModule.forRoot()

MainComponent wordt nu gebootstrapt

4. MainComponent met Routing maken

- Nieuwe component met hoofdmenu en `<router-outlet>`

```
import {Component, OnInit} from '@angular/core';

@Component({
   selector: 'main-component',
   template: `
      <h1>Pick your favorite city</h1>
      <!-- Static 'main menu'. Always visible-->
      <!-- Add routerLink directive. Angular replaces this with correct <a href="..."> -->
      <a routerLink="/home" class="btn btn-primary">List of cities</a>
      <a routerLink="/add" class="btn btn-primary">Add City</a>
      <hr>
      <!-- Dynamically inject views here -->
      <router-outlet></router-outlet>
      <!-- Static footer here. Always visible-->
      `
})
export class MainComponent implements OnInit {
   constructor() {    }
   ngOnInit() { }
}
```

"Hoofdmenu". Let op routerLink

<router-outlet>

Lege Component

5. Eventueel: index.html aanpassen

- Eventueel selector in index.html aanpassen
- Als `MainComponent` een andere selector heeft

```
<div class="container">
   <main-component>
      Loading...
   </main-component>
</div>
```

6. Nieuwe component(en) maken en importeren

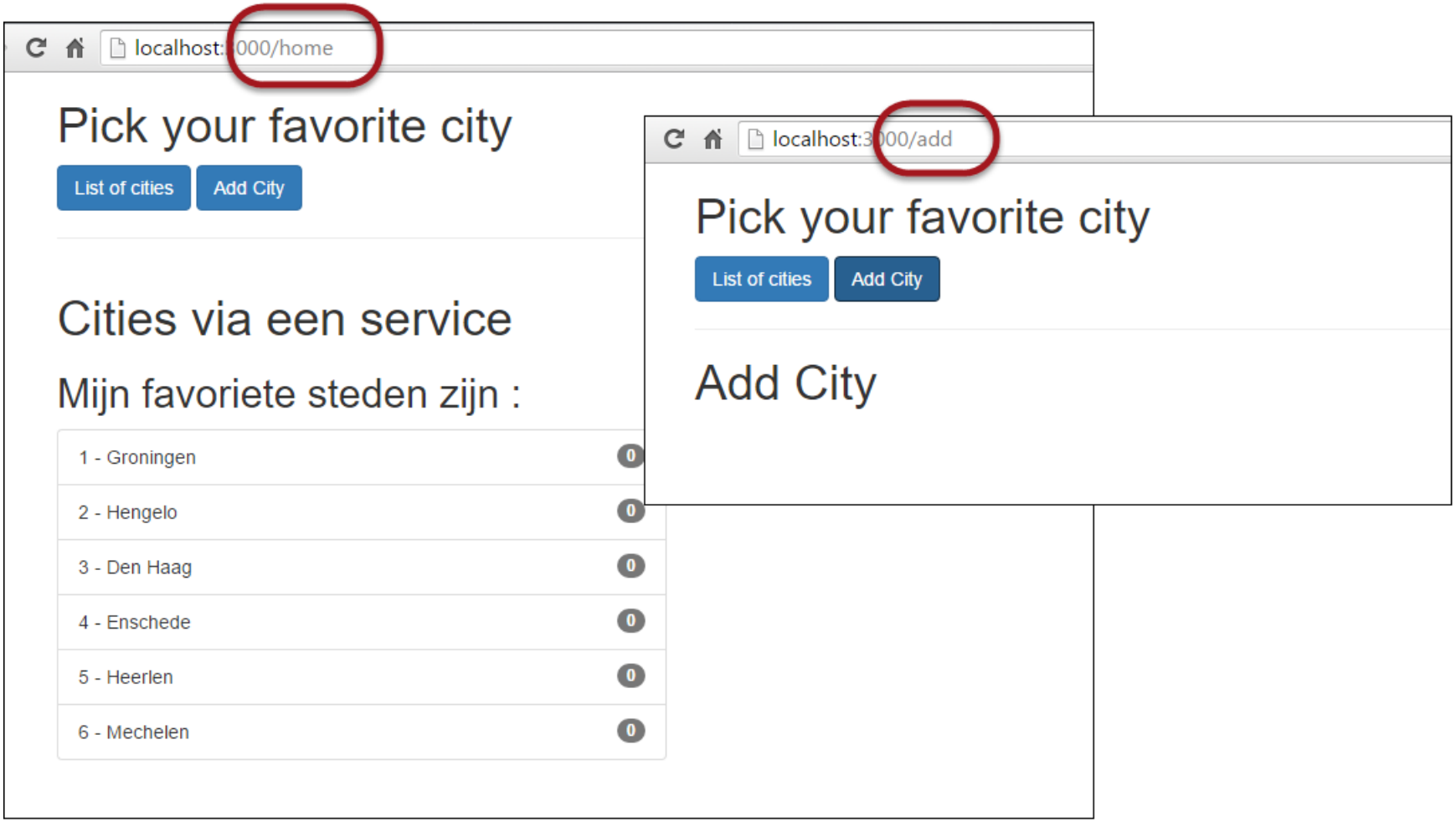
Elke component is een route

```
// city.add.component.ts
import { Component } from ‘@angular/core';

@Component({
   selector: 'a
   template: `<
})

export class Ci
   …
}
```

```
// city.edit.component.ts
import { Component } from ‘@angular/core';

@Component({
   selector: 'edit-ci
   template: `<h1>Edi
})

export class CityEdit
   …
}
```

```
// city.detail.component.ts
import { Component } from ‘@angular/core';

@Component({
   selector: 'detail-city',
   template: `<h1>Detail City</h1> …`
})

export class CityDetailComponent{
   …
}
```

7. Run the application



Catch-all routes

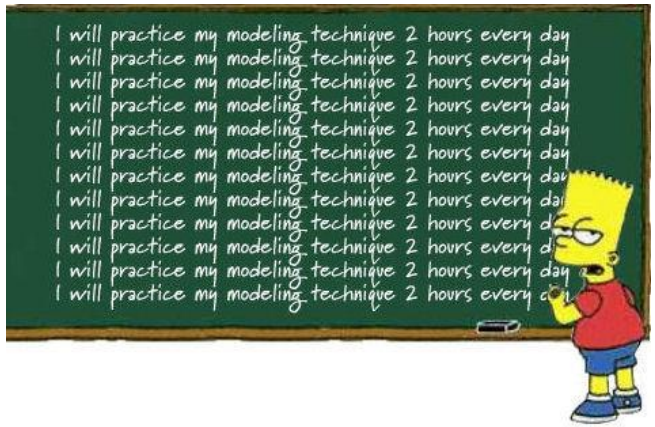
```
6   export const AppRoutes: Routes = [
7       {path: '', component: AppComponent},
8       {path: 'home', component: AppComponent},
9       {path: 'add', component: CityAddComponent},
10      {
11          // catch all route
12          path     : '**',
13          redirectTo: 'home'
14      },
15  ];
16
```

Gebruik `**` voor een catch-all route:

- `Component` opgeven (=route blijft zichtbaar in URL-balk
- `redirectTo`: opgeven (=nieuwe route staat in URL-balk)

Checkpoint

- Routes worden op module-level ingesteld (Angular 1: app-level).
- 1. Eigen applicatie:
  - Volg het stappenplan. Denk aan injecteren van RouterModule, `app.routes.ts` en `<base href="/">` in de HTML
  - = oefening `7a)`
- 2. Voorbeeld: `/400-routing`
  - Voeg een nieuwe component to aan het routing-voorbeeld en zorg dat naar deze component kan worden gerouteerd
- 3. Optioneel: Oefening `7b)`
  - = nieuwe app maken, inclusief `routing`
  - Gebruik de CLI, `ng new <…>`



Kies een workshop….

Routing in Modern Angular

Angular 17+ applicaties hebben een iets andere configuratie.

App.routes.ts en app.config.ts

```
// app.routes.ts -
// Default: EMPTY routing table. Define your own routes.
import { Routes } from '@angular/router';



export const routes: Routes = [];
```

```
// app.config.ts - provide your routes to tye application.
import { ApplicationConfig } from '@angular/core';
import { provideRouter } from '@angular/router';

import { routes } from './app.routes';

export const appConfig: ApplicationConfig = {
  providers: [provideRouter(routes)]
};
```

Main.ts

```
// main.ts - pass in appConfig to configure the application,
// containing the routes you created.
import { bootstrapApplication } from '@angular/platform-browser';
import { appConfig } from './app/app.config';
import { AppComponent } from './app/app.component';

bootstrapApplication(AppComponent, appConfig)
  .catch((err) => console.error(err));
```

Routeparameters

Master-Detail views en –applications

Dynamische routes maken

Doel: Enkele detailpagina voor klanten, producten, diensten, etc.

Leesbare routes als: `/cities/5`, of `products/philips/broodrooster`, enzovoort

Werkwijze:

1. Aanpassen `routing table` in de applicatie.
2. Gebruik `route:ActivatedRoute` in de detail component
3. Schrijf hyperlinks als `<a [routerLink]=…>` met parameter

1. app.routes.ts aanpassen

```
// app.routes.ts
import {Routes} from '@angular/router';
import {AppComponent} from "./app.component";
import {CityAddComponent} from "./city.add.component";
import {CityDetailComponent} from "./city.detail.component";

export const AppRoutes: Routes = [
   {path: '', component: AppComponent},
   {path: 'home', component: AppComponent},
   {path: 'add', component: CityAddComponent},
   {path: 'detail/:id', component: CityDetailComponent}
];
```

2. Detail Component maken

```
// city.detail.component.ts
…
// import {RouteParams} from "@angular/router"; // OLD way
import {ActivatedRoute} from '@angular/router';

@Component({
   selector: 'city-detail',
   template: `<h1>City Detail</h1>
   <h2>Details voor city: {{ id }}</h2>
   `
})

export class CityDetailComponent implements OnInit, OnDestroy {
   id: string;
   currentCity: City;

   constructor(private route: ActivatedRoute) {}

   ngOnInit() {
      this.route.params
         .subscribe((params: any) => {
            this.id = params.id;
         });
   }
}
```

ActivatedRoute

2a. DetailComponent - variants

Using router snapshots

```
// OR:
// Work via Router-snapshot:
// Sometimes we're not interested in future changes of a route parameter.
// All we need the id and once we have it, we can provide the data we want to provide.
// In this case, an Observable can bit a bit of an overkill.
// A *snapshot* is simply a snapshot representation of the activated route.
this.id = this.route.snapshot.params['id'];
this.name = this.route.snapshot.params['name'];
```

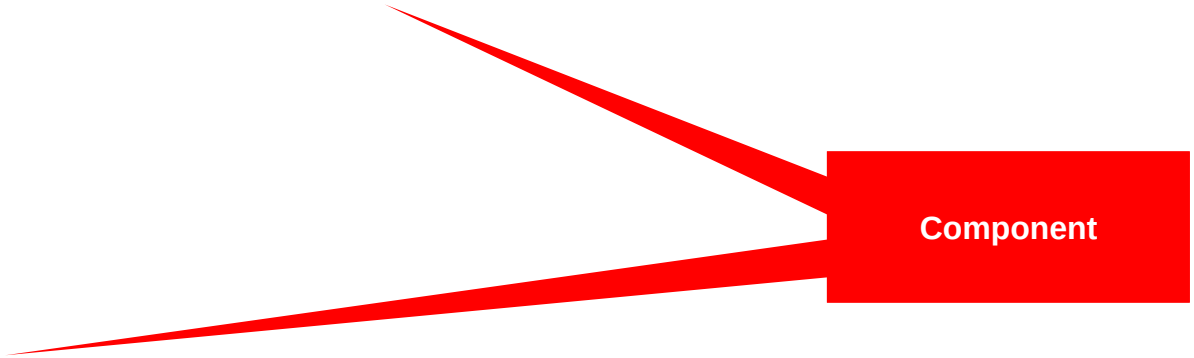
2b. DetailComponent - variants

```
ngOnInit() {
   // NEW:
   this.sub = this.route.params
      .subscribe((params: any) => {
         this.id = params['id'];
         this.name = params['name'];
      });
}

ngOnDestroy() {
   // If subscribed, we must unsubscribe before Angular destroys the component.
   // Failure to do so could create a memory leak.
   this.sub.unsubscribe();
}
```

.unsubscribe()

3. Detail component toevoegen aan Module

```
// app.module.ts
…
// Components
import {CityDetailComponent}  from './city.detail.component';

@NgModule({
   imports     : [
      …
   ],
   declarations: [
      …
      CityDetailComponent
   ],
   providers   : [CityService],
   bootstrap   : [MainComponent]
})
export class AppModule {
}
```



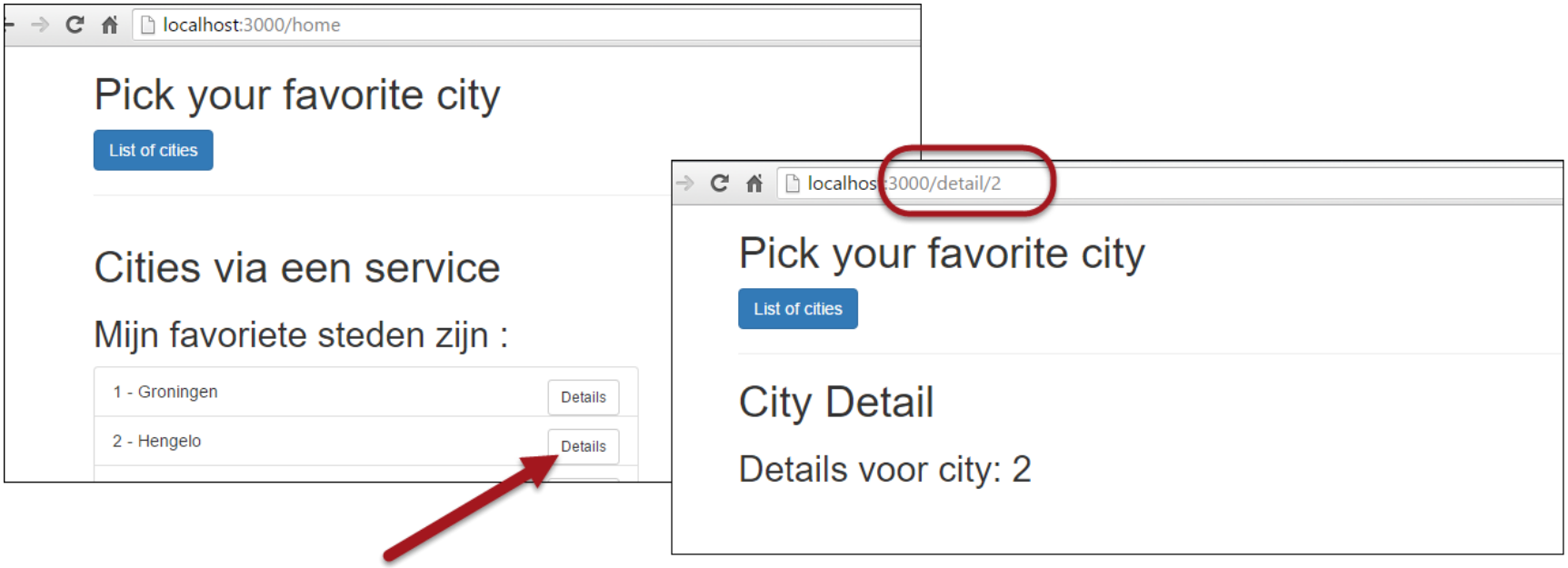
4. App Component (‘Master View’) aanpassen

```
<li *ngFor="let city of cities" class="list-group-item">
   <a [routerLink]="['/detail', city.id]">
      {{ city.id}} - {{ city.name }}
   </a>
</li>
```

Let er op dat `[routerLink]` nu dynamisch moet worden gevuld en dus binnen `[…]` moet staan voor attribute binding

Meegeven van parameters

- Let op meegeven van *array van parameters* aan `[routerLink]`
- Parameters worden gematched op positie. Niet op naam.
- Optioneel : service uitbreiden om specifiek product/item te retourneren



Optionele parameters

Optionele routeparameters meegeven op de querystring

Optionele parameters : [queryParams]

In de HTML

```
<a [routerLink]="['/detail', city.id, city.name]"
   [queryParams]="{province:city.province, population:180000}">
    {{ city.id}} - {{ city.name }}
</a>
```

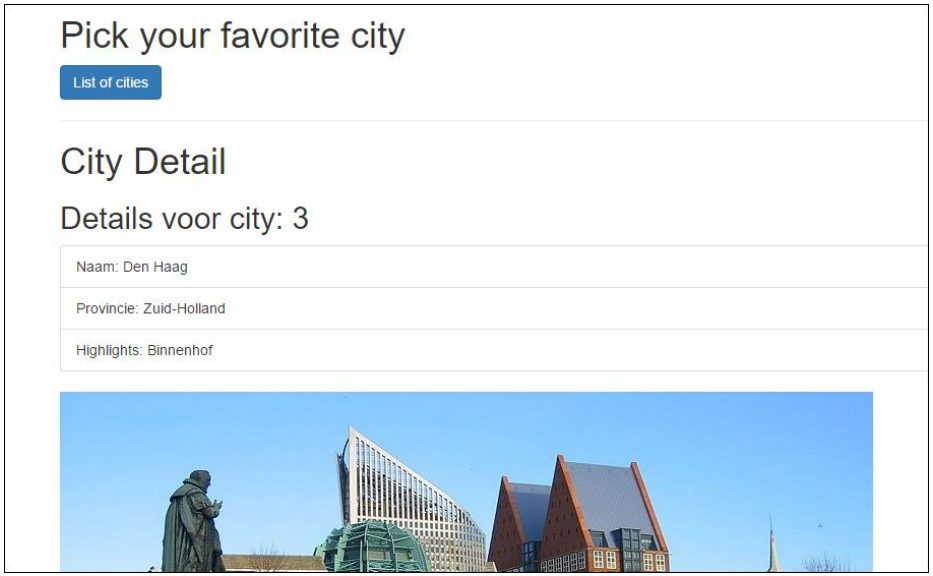
In de class

```
this.route.queryParams.subscribe((params: any) => {
    this.province = params.province;
})
```

Vervolg – details via Service

- Uncomment de regels die te maken hebben met cityService:

```
// NEW, with fetching details via Service:
this.sub = this.route.params
    .map(params => params['id'])
    .switchMap(id => this.cityService.getCity(id))
    .subscribe((city) => {
       this.currentCity = <City>city[0];
    });
```



In city.service.ts:

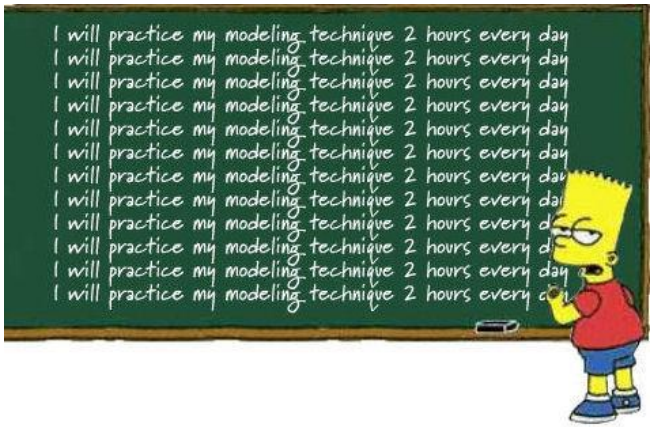
- Bijvoorbeeld (kan beter, maar het werkt wel):

```
// retourneer een city, op basis van ID
getCity(id: string): Observable<City[]> {
   return this.http.get('app/cities.json’)
         .pipe(
                 map(cities => cities.filter((city: City) => {
                 return city.id === parseInt(id);
               })
         )
}
```

Checkpoint

- RouteParameters worden met `:parameterName` ingesteld in `app.routes.ts`.
- Denk aan injection van `ActivatedRoute` in de component.
- Hierin is een property `.params` aanwezig met de meegegeven parameters.
- `<a [routerLink]="…">` gaan uitbreiden
- Voorbeeld: `\401-router-parameter`
- Oefening: `7c)`

Oefening….



More on routing

- Router Guards – delen van je routes beveiligen
- Child Routes
- Named Router Outlets
  - http://onehungrymind.com/named-router-outlets-in-angular-2/
- Router resolvers
  - https://blog.thoughtram.io/angular/2016/10/10/resolving-route-data-in-angular-2.html
- Lazy Loading – Applicatie opdelen in Modules en laden *on demand*
  - https://angular.io/guide/router#lazy-loading-route-configuration

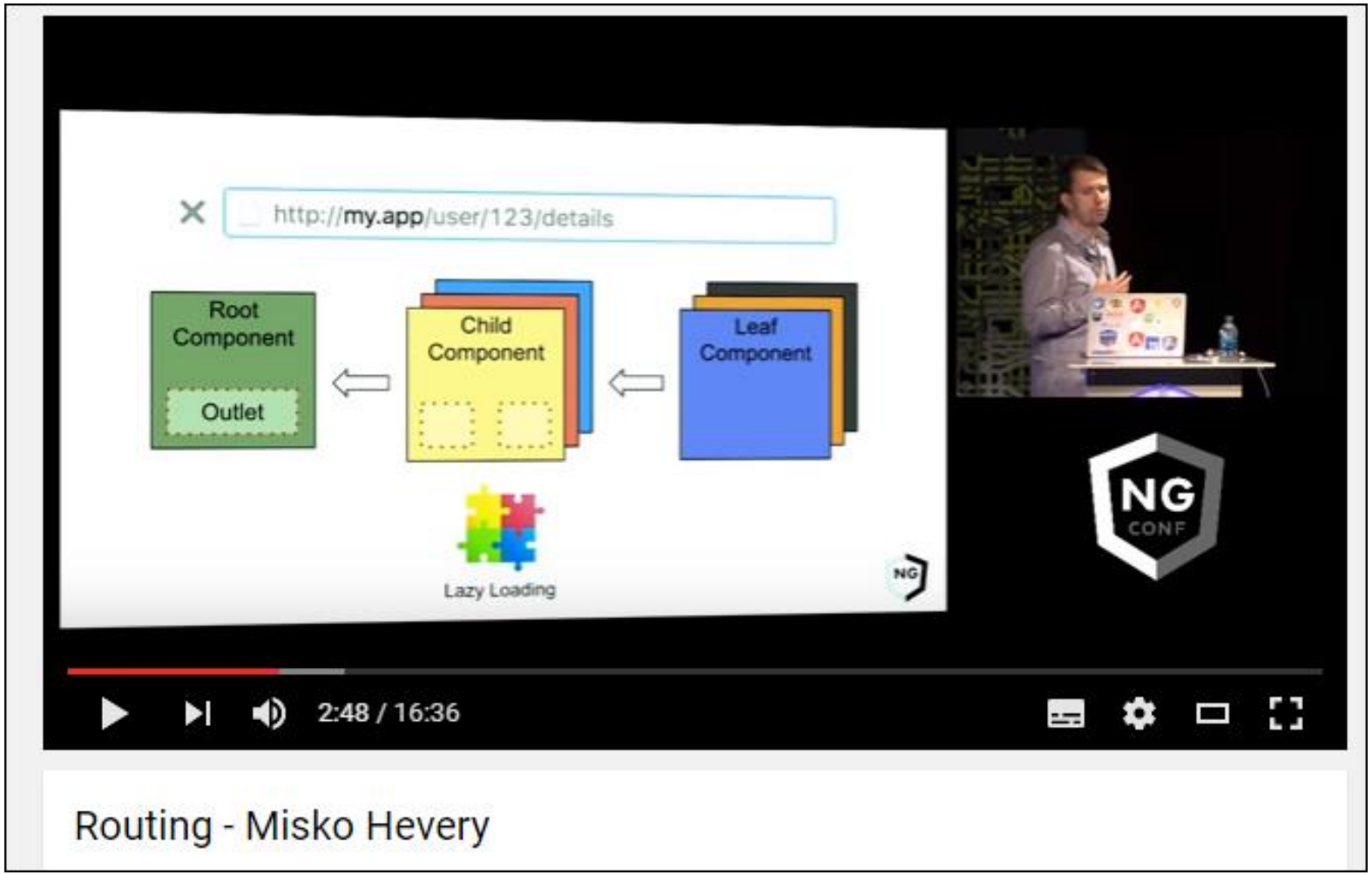
More info

More background information on routing

Achtergrondinfo

- https://angular.io/docs/ts/latest/guide/router.html
- http://blog.thoughtram.io/angular/2016/06/14/routing-in-angular-2-revisited.html
- http://blog.thoughtram.io/angular/2016/07/18/guards-in-angular-2.html
- https://vsavkin.com/
- https://angular-2-training-book.rangle.io/handout/routing/child_routes.html

New Component Router

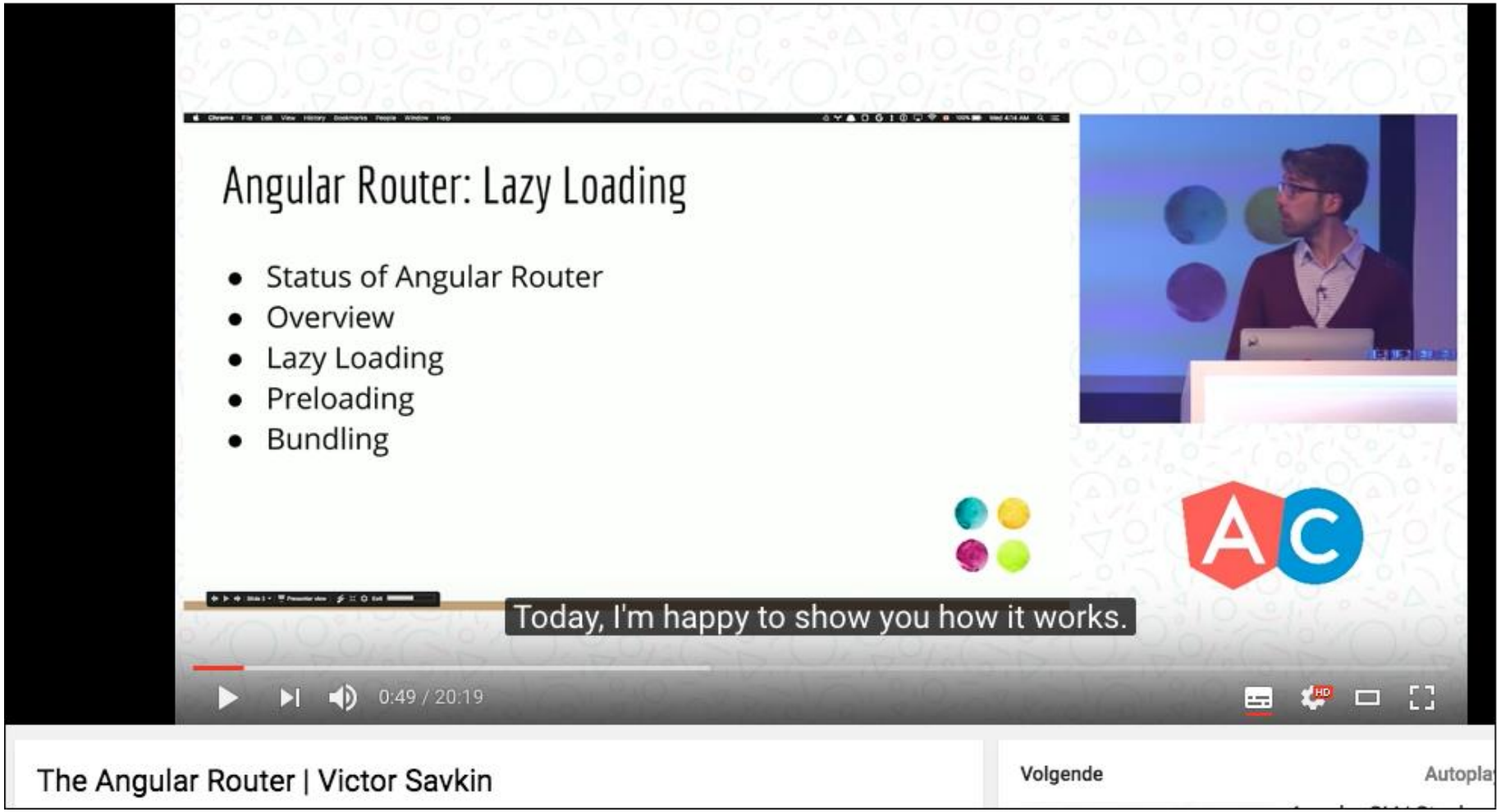


https://www.youtube.com/watch?v=d8yAdeshpcw

Victor Savkin (=maker van de router)

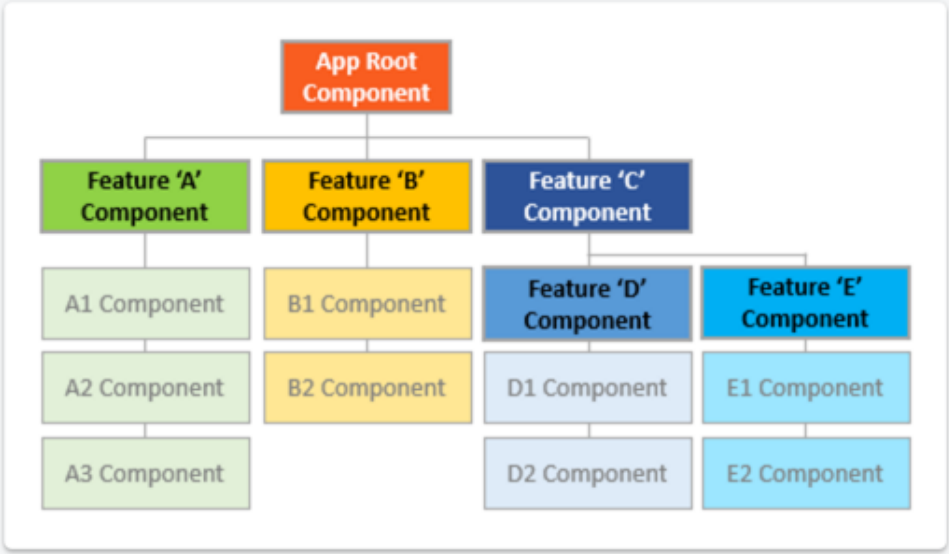
https://leanpub.com/router

https://www.youtube.com/watch?v=QLns6s02O48



Advanced routing

It's looking good as a general pattern for Angular applications.



- each feature area in its own module folder
- each area with its own root component
- each area root component with its own router-outlet and child routes
- area routes rarely (if ever) cross

https://angular.io/docs/ts/latest/guide/router.html

Victor Savkin on Routing

Victor Savkin – creator of the router

https://vsavkin.com/